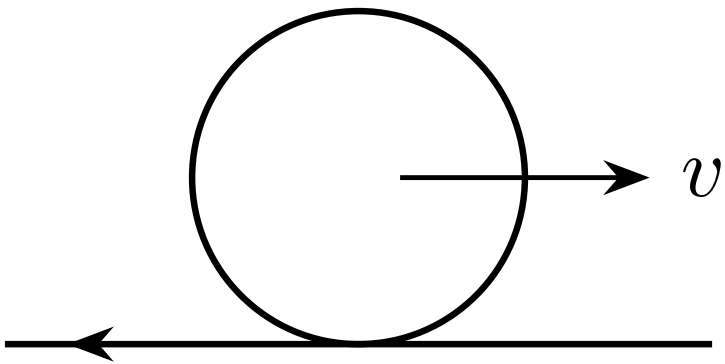
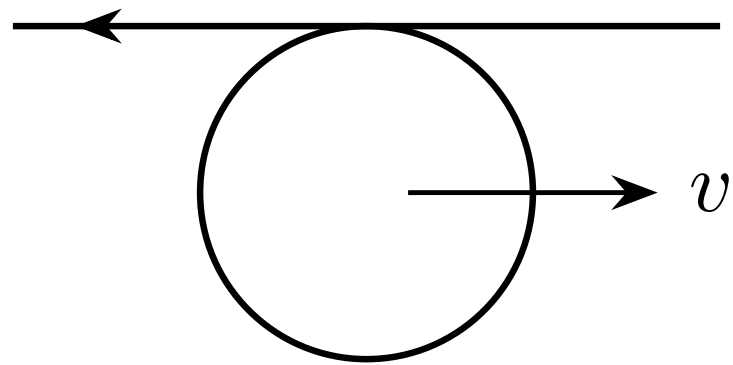
下板



$$u_- = v - r\omega$$

上板



$$u_+ = v + r\omega$$

同样的切向冲量，上、下接触的角冲量符号相反